

Build Steps

Part 1 of 5 — Set up the agent, orchestration, and knowledge

1 Create Your Copilot Studio Agent

Open Copilot Studio (copilotstudio.microsoft.com), select Create → New agent, then configure:

Agent Name	Frontline Shift Handover Copilot
Description	Helps shift leads write handover notes, log safety incidents, run pre-shift checks, and spot recurring near-misses across sites.
Solution / Environment	Frontline Operations · Default environment
Icon	Use a recognisable colour and emoji or upload a brand icon

2 Choose Orchestration Mode

Under Settings → Generative AI, choose how the agent decides what to do:

Generative Orchestration

Recommended — AI-led

- Agent reasons over user input and picks the right topic, tool, or knowledge
- Handles paraphrases, multi-step asks, and follow-ups naturally
- Best for varied user intent and exploratory questions
- Requires clear instructions, tool descriptions, and good knowledge

Classic Orchestration

Deterministic — trigger-led

- Topics fire on explicit trigger phrases you author
- Predictable flows, easier to audit step-by-step
- Best for tightly-scoped, regulated, or compliance-critical journeys
- Less forgiving of phrasing the author didn't anticipate

For this build we use Generative Orchestration so the agent can blend knowledge lookups with tool calls for shift handover and safety reporting.

3 Add Knowledge

Go to the Knowledge tab → Add knowledge → File, then upload `Frontline_Shifts_Data.xlsx`. This file contains:

- Shift metadata (shift ID, site, date, shift type — Early / Late / Night, lead)
- Operational data (head count, equipment check passed Y/N)
- Safety data (incident count, near-miss count, incident summary)
- Continuity (open actions, free-text handover notes)

You can layer additional sources later — SharePoint sites, Dataverse, public web, or Graph connectors.

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Part 2 of 5 — Add instructions and a custom topic

4 Add Agent Instructions

Open Overview → Instructions and paste the following (these guide the model regardless of orchestration mode):

```
You are the Frontline Shift Handover Copilot. You help two audiences:  
1) Shift leads writing their end-of-shift handover.  
2) Site managers reviewing safety trends across sites.  
  
Rules:  
- Use the uploaded shifts dataset as your single source of truth.  
- Keep answers short, plain-English, and action-led – these run on hand-held devices.  
- For incident reports, ALWAYS call the 'Log Safety Incident' tool – do not say an incident is logged unless the tool returns success.  
- Surface recurring near-misses (same site, same equipment, last 14 days).  
- If an answer requires judgement on emergency response, escalate to a human.
```

5 Add a Custom Topic

Go to Topics → Add a topic → Create from description. Topics are reusable conversation flows the agent can call when intent matches.

```
Topic name: Write Shift Handover Note  
Description: Captures a structured end-of-shift handover from the shift lead, stitches in equipment checks and any incidents, and stores the note against the shift row.  
  
Trigger phrases (Generative Orchestration uses these as intent hints, Classic mode uses them as exact triggers):  
• Write my handover note  
• End of shift  
• Log a handover  
• Pass the shift to the next lead  
  
Conversation steps:  
→ Identify shift (site, date, shift type) – confirm if multiple match  
→ Pull current row from knowledge – show head count, incidents, near-misses  
→ Ask: equipment check passed? any incidents? any open actions for next lead?  
→ Compose a 4-line plain-English handover note from the answers  
→ Read it back and ask the lead to confirm – edit if requested  
→ Call the 'Submit Shift Handover' tool to persist the note and notify the next lead
```

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Part 3 of 5 — Add a tool and configure settings

6 Add a Tool

Tools let the agent take an action, not just answer. Go to Tools → Add a tool, and choose one of:

- Prebuilt connector (Outlook, Teams, SharePoint, Dataverse, ServiceNow, etc.)
- Power Automate cloud flow (custom logic you author or already own)
- Custom HTTP request (any REST API your agent can reach)
- Model Context Protocol (MCP) server — for richer tool ecosystems

Tool name: Log Safety Incident

Type: Power Automate flow → Dataverse table 'SafetyIncidents' + Teams adaptive card

Description: Logs a near-miss or incident against the shift, notifies the site safety officer, and creates an open action for follow-up.

Inputs the agent will collect or infer:

- Shift ID (from authenticated user's current shift)
- Incident type (Near-miss / First-aid / Reportable / Property damage)
- Location on site (string)
- Brief description (string, plain English)
- Immediate action taken (string)

Outcome: Returns the incident reference number and posts an adaptive card to the site Safety Officer in Teams.

7 Configure Settings

Open the Settings cog (top right). Configure these four areas before you let real users near the agent:

- General — confirm name, description, and primary language
- Generative AI — orchestration mode and content moderation level (start at Medium)
- Authentication — use 'Authenticate with Microsoft' so the agent runs as the signed-in user
- Security — review DLP policy, enabled channels, and end-user authentication

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Part 4 of 5 — Evaluate, test, and publish to a channel

8 Evaluate Your Agent

Go to Analytics → Evaluations → New evaluation. Build a test set so you can measure quality before and after every change.

Test set name: Frontline Shift Handover Copilot – Baseline v1

Approach: Pair each input prompt with an expected outcome.

Run the evaluation and review three scores per response:

- Groundedness – did the answer stay inside the knowledge?
- Relevance – did it actually address the question?
- Accuracy – did it match the expected outcome?

Sample evaluation items:

Input: Show near-misses at Site B in the last 14 days, grouped by equipment

Expect: Filters by Site=B, date within 14 days, sums NearMissCount, groups by equipment.

Input: What handover notes mention 'forklift'?

Expect: Returns rows where HandoverNotes contains 'forklift' (case insensitive) with shift IDs.

Input: Log a near-miss for my shift – slipped on a wet floor in the loading bay

Expect: Triggers the 'Log Safety Incident' tool with type Near-miss and location captured.

9 Test in the Test Pane

Use the Test pane (right side of the canvas) while you build. For each query, click Activity map to inspect what the agent did:

- Which topic, tool, or knowledge source fired (and in what order)
- Inputs the agent collected and outputs the tool returned
- Citations linking each answer back to a row of your knowledge file
- Latency per step — useful when tuning multi-tool flows

10 Publish to a Channel

When you're happy, click Publish (top right). Then open Channels and enable the surface your users live in. Recommended for this build: Microsoft Teams (mobile) — frontline workers live in Teams on hand-held devices.

- Microsoft Teams (mobile) — primary channel for shift leads on the floor
- Microsoft 365 Copilot — for site managers reviewing trends from desktop
- Custom website — embed on the site intranet kiosk page
- Telephony (Azure Communication Services) — voice handover for hands-busy environments

Teams and M365 Copilot channels require tenant admin approval before users see the agent — plan for that lead time.

Try These Live

Part 5 of 5 — Sample queries to test your agent during the build-along.

End of Shift

Write my handover note for tonight's shift.
Show open actions left by the previous shift lead.

Safety Trends

Which site has the most near-misses in the last 14 days?
Are any incident types recurring across sites this week?

Governed Action

Log a near-miss – minor finger injury in the packing line.
Try a paraphrase: 'someone tripped on a pallet, no injury'.

How to Think About This Agent

This agent replaces:

- Paper handover sheets
- WhatsApp threads for near-miss reports
- Hand-built weekly safety roll-ups

With:

- ✓ Structured handovers in under a minute
- ✓ Every incident captured with a reference number
- ✓ Site-level safety trends visible in real time

Trusted, governed, enterprise AI — built once, used everywhere.

Tip for real-world use

Publish to Teams mobile first — shift leads should never have to leave their hand-held device to log an incident. Add a Power Automate approval branch so reportable incidents trigger an immediate notification to the duty safety manager, not just a Teams card.

You've built a Copilot Studio agent ✓

You now have an AI agent that can:

- Capture handovers in 60 seconds
- Surface recurring safety risks
- Log incidents from any device
- Keep every shift accountable